Finding the equation of a parabola

1 A parabola of the form $y = ax^2$ goes through the point $(2, -4)$.

a What is the value of a?

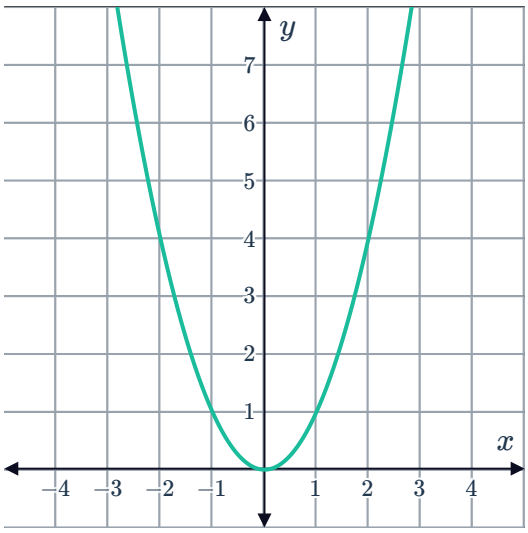
b What are the coordinates of the vertex?

c Sketch the graph of the parabola.

2 Consider the graph of $y = x^2$:

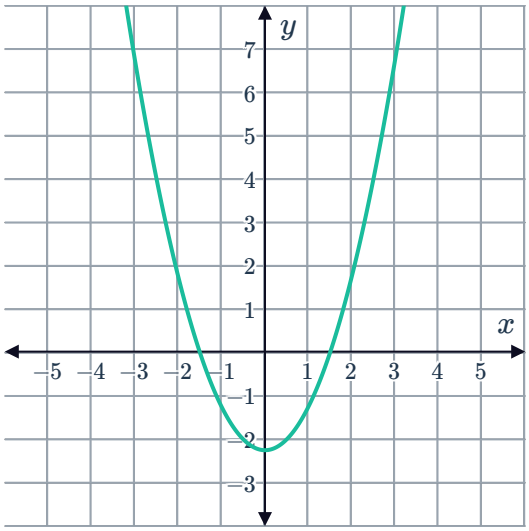
a Sketch the graph of the parabola that has the same vertex as $y = x^2$ and that passes through $(2, 2)$.

b What is the equation of the new parabola?

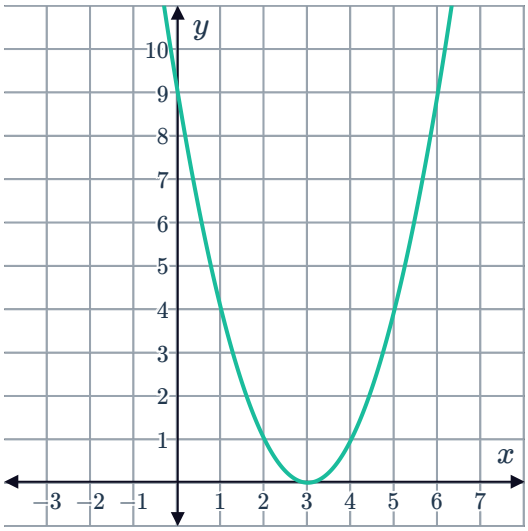


3 Determine whether the following graphs represent a parabola of the form $y = (x - a)^2$:

a

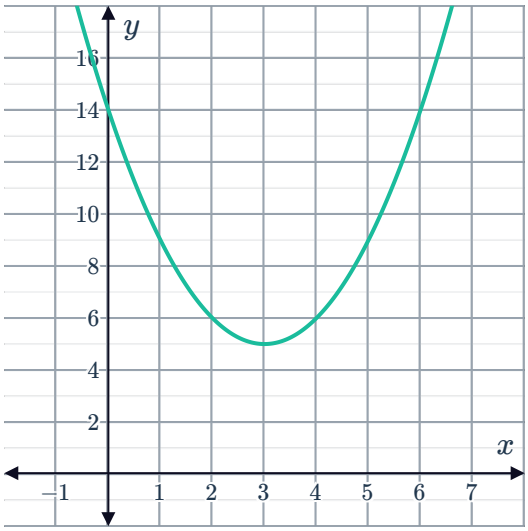


b



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4 Consider the following graph:



a Determine the coordinates of the vertex of the parabola.

b Determine the equation of the parabola of the form $y = a(x-h)^2 + k$.

5 A parabola has its turning point at $x = -3$ and one of the x-intercepts is at $x = 1$.

a What is the other x-intercept?

b If the parabola has a y-intercept at 3, write down the equation of the parabola.

c What are the coordinates of the turning point?

6 Write down the equation of the parabola that has the same shape as $f(x) = 6x^2$, and vertex at $(-2, 3)$.

7 A parabola is of the form $y = (x-h)^2 + k$. It has x-intercepts at $(1, 0)$ and $(-5, 0)$.

a Determine the equation of the axis of symmetry.

b Hence or otherwise find the equation of the curve.

8 Consider the equation of a parabola whose x-intercepts are -10 and 4, and whose y-intercept is -40. Find the equation of the parabola.

9 Consider the equation $y = (x-b)(x+b)$.

a Expand and simplify $(x-b)(x+b)$.

b Hence, find the value of a such that the parabola $y = x^2 + ax - 9$ will have x-intercepts that are equidistant from the origin.

10 A parabola has equation of the form $y = (x-a)(x-b)$.

a The parabola has x-intercepts at $x = -1$ and $x = -5$. Write down its equation.

b What is the y-intercept of the parabola?

c Sketch the graph of the curve.

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11 A quadratic function is of the form $y = (x - a)(x - b)$.

a The corresponding parabola has x-intercepts $x = \sqrt{3}$ and $x = -\sqrt{3}$. Write down its equation in expanded form.

b What is the y-value of the point on the parabola where $x = 1$?

c Sketch the graph of the curve.

12 Consider two different parabolas:

- Parabola 1: has two x-intercepts $(-6, 0)$ and $(4, 0)$
- Parabola 2: has a vertex at $(-1, 3)$

a How many unique parabolas have the same intercepts as Parabola 1?

b How many unique parabolas have the same vertex as Parabola 2?

c How many unique parabolas have the intercepts as Parabola 1 and the same vertex as Parabola 2?

d The equation of the parabola from part (c) can be written as $y = a(x + 6)(x - 4)$. Find the value of a and hence write the equation of the parabola in expanded form.

13 Find the equation of the quadratic function that has a vertex at $(-12, 3)$ and that passes through the point $(-4, 19)$.

14 The parabola $y = 2x^2 + bx + 1$ has its axis of symmetry at $x = -1$. Find the value of b.

15 Determine the value of c if the parabola $y = x^2 + 4x + c$ has exactly one x-intercept.

16 Determine the value of k such that $y = (k + 5)x^2 + (k + 5)x + k + 1$ touches the x-axis.

17 We want to find the quadratic function $y = ax^2 + bx + c$ whose graph passes through the points $(1, 3)$, $(3, -1)$, and $(4, 0)$.

a Substitute each point into the quadratic function, to form three equations in terms of a, b, and c.

b Find the value of c.

c Find the value of a.

d Hence, find the value of b.

e Hence, state the equation of the quadratic function that passes through these points.

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Maximum and minimum values

18 Consider the parabola $y = -x^2 - 4$.

- **a** Does it have a minimum or maximum y-value?
- **b** What are the coordinates of the vertex?

19 Consider the parabola $y = x^2 - 18x + 10$.

- **a** Is the graph of the function concave up or down?
- **b** State the value of x at which the minimum value of the function occurs.
- **c** State the minimum y-value of the function.

20 Find the maximum value of y for the quadratic function $y = -x^2 + 10x - 25$.

21 Consider the quadratic function $f(x) = -7.5x^2 - 1.3x - 1.3$.

- **a** Find the x-coordinate of the vertex correct to three decimal places.
- **b** Hence, find the maximum value obtained by the function, correct to two decimal places.

22 The minimum value of the function $y = 9x^2 + 108x + m$ is $y = 6$.

- **a** Find the x-coordinate of the minimum point.
- **b** Find the value of m.

23 The maximum value of the function $y = -7x^2 - 140x + m$ is $y = 3$.

- **a** Find the x-coordinate of the maximum point.
- **b** Find the value of m.

Graphing quadratics using technology

24 A parabola of the form $y = (x - h)^2 + k$ is symmetrical about the line $x = 2$, and its vertex lies 3 units above the x-axis. If we are to sketch this curve on a graphing calculator, what equation do we type into our graphing calculator?

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25 Using a graphing calculator, or other technology, sketch the quadratic function $y = x^2 + 4x - 5$.

- **a** Is the parabola is concave up or down?
- **b** Find the x-intercepts of the function.
- **c** Find the y-intercept.
- **d** Determine the axis of symmetry.
- **e** Hence or otherwise find the vertex of the curve.

26 Using a graphing calculator, or other technology, find the minimum value of y for the quadratic function:

$$y = x^2 + x - 20$$

27 Using a graphing calculator, or other technology, sketch the curve $y = x^2 + 6.2x - 7$.

- **a** Determine the axis of symmetry.
- **b** Determine the minimum value of y.

28 Use your calculator, or other technology, to graph $y = 3x^2 + 18x + 36$.

- **a** Is the graph concave up or concave down?
- **b** What is the minimum y-value?
- **c** What is the x-value corresponding to the minimum y-value?
- **d** For what values of x is the parabola decreasing?

29 Use your calculator, or other technology, to graph $y = -5x^2 - 30x - 51$.

- **a** Is the graph concave up or concave down?
- **b** What is the maximum y-value?
- **c** What is the x-value corresponding to the maximum y-value?
- **d** For what values of x is the parabola increasing?

30 Use your calculator, or other technology, to graph $y = 4x^2 + 48x + 135$.

- **a** What is the vertex of the graph?
- **b** What are the coordinates of the y-intercept?

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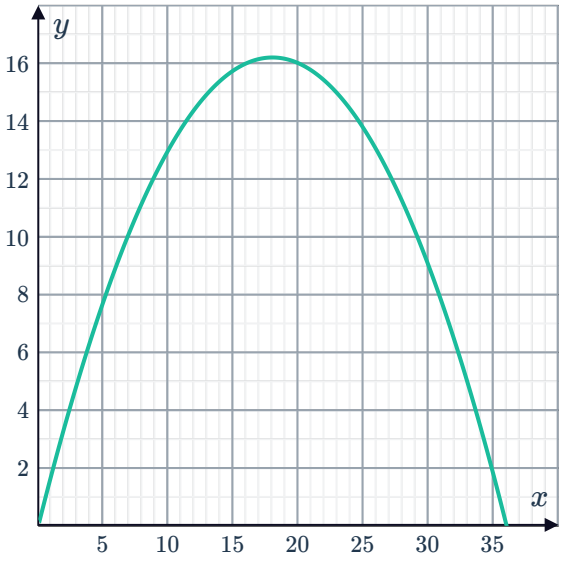
Applications

31 A skydiving instructor wants to use the equation $y = at^2 + c$ to model the height in metres of a skydiver above the ground t seconds after jumping out of the plane.

- **a** What signs should a and c be?
- **b** She does a test run, jumping out of the plane at a height of 2560 metres. Find the value of c.
- **c** After 160 seconds, the instructor lands on the ground. Find the value of a using an equation.
- **d** After 23 seconds, how far above the ground was she?

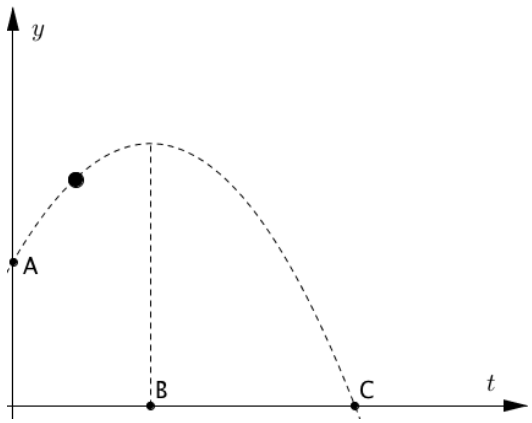
32 Large sprinklers are used to water crops. When water is sprayed, it takes the path of a parabola. The stream of water reaches a greatest height of 16.2 metres above the ground, 18 metres from the sprinkler.
Let the position of the sprinkler be $(0, 0)$. Let x be the horizontal distance from the sprinkler and y be the height of the water above the ground.

- **a** The graph of the water from the sprinkler is shown. Write down the rule linking x and y.
- **b** If the sprinkler were to rotate 360°, what area of crops would it irrigate? Give your answer to two decimal places.



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33 In a game of baseball, the ball is hit high up into the air. Initially the ball is struck 2.5 metres above the ground and hits the ground 6 seconds later. It reaches its greatest height 2 seconds after being hit.



a Determine the coordinates of the points labelled A, B and C in the diagram.

b Using the form of the parabola $y = a(t-h)^2 + k$, where t is the number of seconds after the ball is hit and y is the height of the ball above the ground, determine the value of h.

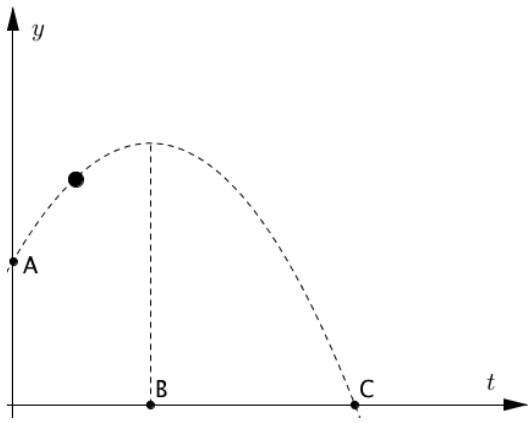
c By substituting $t = 0$ and $y = 2.5$ into the equation, form an equation in terms of a and k.

d By substituting $t = 6$ and $y = 0$ into the equation, form another equation in terms of a and k.

e Solve for a.

f Hence find k, the greatest height reached by the ball.

34 In a game of tennis, the ball is hit high up into the air. Initially the ball is struck 3.5 metres above the ground and hits the ground 7 seconds later. It reaches its greatest height 3 seconds after being hit.



a Determine the coordinates of the points labelled A, B and C in the diagram.

b Using the form of the parabola $y = a(t-h)^2 + k$, where t is the number of seconds after the ball is hit and y is the height of the ball above the ground, determine the value of h.

c By substituting $t = 0$ and $y = 3.5$ into the equation, form an equation in terms of a and k.

d By substituting $t = 7$ and $y = 0$ into the equation, form another equation in terms of a and k.

e Solve for a.

f Hence, find k, the greatest height reached by the ball.

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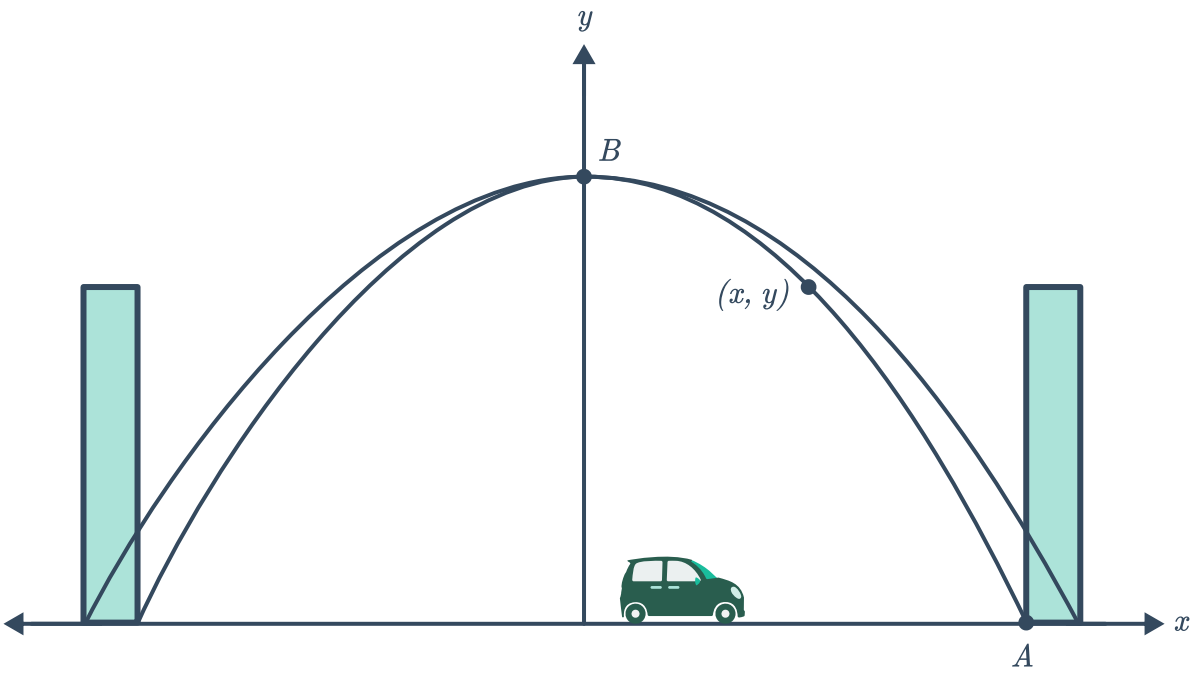
35 To minimise the chance of large avalanches, small explosives are fired onto mountain sides. One such explosive is fired at a height of 24 metres and follows a parabolic flight path. It reaches a maximum height of 122 metres, 180 metres horizontally from where it is fired. Let x and y represent the horizontal and vertical distances respectively, of the explosive from where it is fired.

a Assuming the position where it is fired is $(0, 0)$, determine the vertex of the parabolic path.

b Using the form of the parabola $y = -a(x - h)^2 + k$, determine the exact value of a. Then use this value to determine the equation between x and y.

c The explosive hits a slope that is 361 metres horizontally from the point where it was fired. Does the explosive land above, at the same height or below the point where it was fired?

36 The arch of a parabolic bridge starts 18 metres to the left of its centre and ends 18 metres to the right of its centre, and the peak of its arch is 108 metres above the road:



a Determine the coordinates of the points labelled A and B on the diagram.

b Find a formula for y, the height of the arch above the road, at distance x metres from the centre of the bridge.

c Pedestrians can climb along the arch up to a vertical point that is half the height of the arch. At their highest point, how far horizontally are pedestrians from the centre of the arch?

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37 Vincent is training for a remote control plane aerobatics competition. He wants to fly the plane along the path of a parabola, and so has chosen the equation $y = 3x^2$, where y is the height in metres of the plane from the ground, and x is the horizontal distance in metres of the plane from its starting point.

a Complete the following table of values of the height and distance.

x	0	1	2	3	4
y					

b Plot the shape of the path of the plane.

c What is the lowest height of the plane?

d What x-value corresponds to this minimum y-value?

e What are the coordinates of the vertex?

38 An object is released 700 metres above ground and falls freely. The distance the object is from the ground is modelled by the formula $d = 700 - 16t^2$, where d is the distance in metres that the object falls and t is the time elapsed in seconds. This equation is graphed below:

a What does it mean that when $t = 0$, we get $d = 700$?

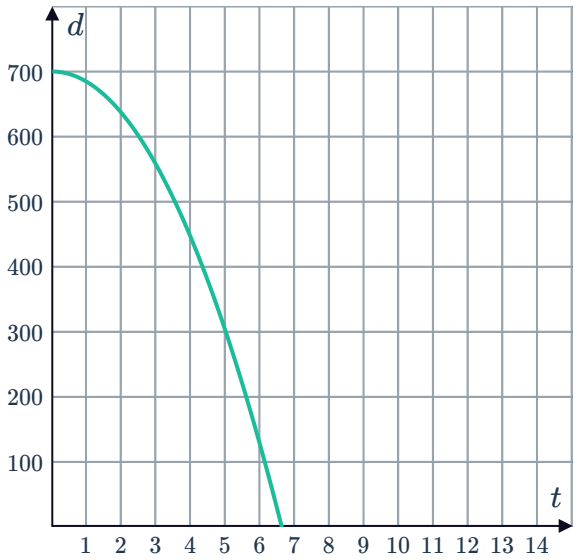
b Is the curve increasing or decreasing?

c Determine whether the following statements are true for the object.

i The object only loses height until it hits the ground.

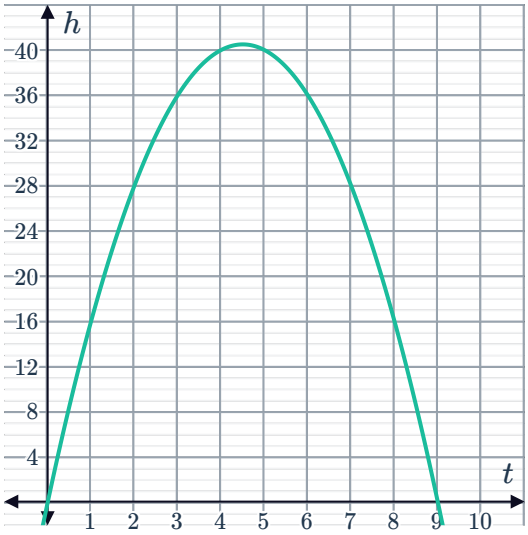
ii The object falls faster and faster until it hits the ground.

iii The object gains height until it reaches 700 metres above ground.



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39 Beth throws a pebble vertically upwards. After t seconds its height h metres above the ground is given by the formula $h = 18t - 2t^2$. This function has been graphed below:



a State the meaning of the point $(4.5, 40.5)$.

b What are the values of t where the graph of h intercepts the horizontal axis?

c What do the intercepts you found in part (b) represent in context?

40 A rectangle has width w, height $20 - w$, area A, and perimeter 40. The equation for the area of the rectangle is $A = w\,(20 - w)$.

a Sketch the graph of A plotted against the width w.

b Find the values of w where the graph intercepts the horizontal axis?

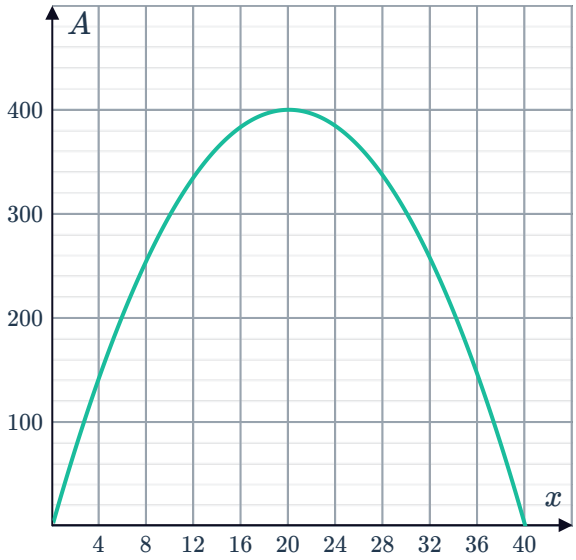
c What can be said about the rectangle when $w = 20$?

d What is the largest possible value of A?

41 A rectangle is to be constructed with 80 metres of wire. The rectangle will have an area of

$$A = 40x - x^2$$

where x is the length of one side of the rectangle. The graph of this function is given:



a What are the x-intercepts of this graph?

b What is the x-value of the vertex?

c What does the height of the parabola at $x = 20$ represent?

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42 When an object is thrown into the air, its height above the ground is given by the equation

$$h = 193 + 24s - s^2$$

where s is its horizontal distance from where it was thrown.

a Find the value of s, at the point when it reaches its greatest height above the ground.

b Find the maximum height reached by the object.

43 A football is kicked into the air and its height h metres above the ground at time t seconds after being kicked is given by $h = -t^2 + 20t$.

a Assuming the ball starts at height 0, at what time, t, will it hit the ground?

b What is the greatest height the ball reaches above the ground?

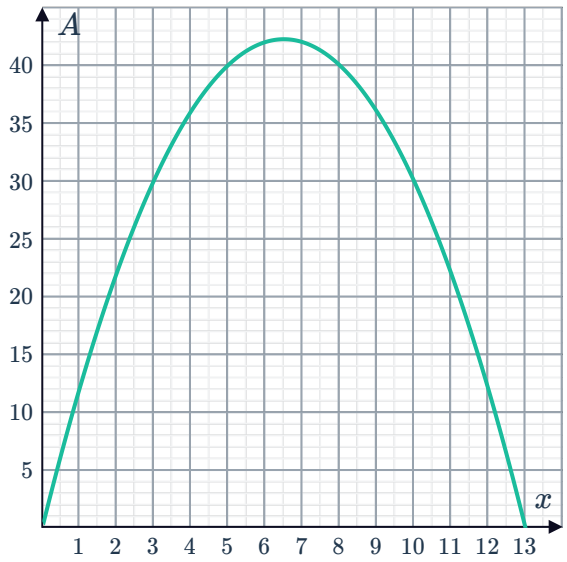
44 A rectangular enclosure is to be built for an animal. Zookeepers have 26 metres of fencing, but they want to maximise the area of the enclosure. Let x be the width of the enclosure. The area function $A = x\,(13 - x)$ has been graphed below:

a Form an expression for the length of the enclosure in terms of x.

b Form an expression for A, the area of the enclosure.

c What width will allow for the greatest possible area?

d What is the greatest possible area of such an enclosure?



Applications using technology

45 The Millau Viaduct in Millau, Southern France, is 270 metres tall.
We can model the behaviour of objects falling from the viaduct using Galileo's formula for falling objects: $d = 4.9t^2$, where d is distance fallen in metres and t is time in seconds since the object was dropped.
Use technology to graph this equation and then answer the following questions:

a What is the vertex of the graph?

b What does the point at $(0, 0)$ represent?

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46 The kinetic energy E of a moving object is given by $E = \frac{1}{2}mv^2$, where m is its mass in kilograms and v is its speed in metres/second.
Graph this equation for a vehicle with a mass of $1400\ \text{kg}$ using a calculator or other technology. What is the speed when the kinetic energy is $137\,200\ \text{j}$?

47 An object launched from the ground has a height (in metres) after t seconds that is modelled by the equation $y = -4.9t^2 + 58.8t$. Graph this equation using a calculator or other technology then answer the following questions.

- **a** What is the maximum height of the object?
- **b** After how many seconds is the object at its maximum height?
- **c** After how many seconds does the object return to the ground?

48 An object is released 900 metres above ground and falls freely. The distance the object is from the ground is modelled by the formula $d = 900 - 4.9t^2$, where d is the distance in metres that the object falls and t is the time elapsed in seconds.
Graph this model using a calculator or other technology and answer the questions below.

- **a** What is d when $t = 0$?
- **b** What does the answer to part (a) mean in context?
- **c** What is the value of t when $d = 0$? Round the answer to two decimal places.
- **d** What does the answer to part (c) mean in context?

49 A rectangle is to be constructed with 60 metres of wire. The rectangle will have an area of

$$A = 30x - x^2$$

where x is the length of one side of the rectangle.
Graph this equation using a calculator or other technology then answer the questions below.

- **a** What are the x-intercepts?
- **b** What is the x-coordinate of the vertex?
- **c** What does the height of the parabola at $x = 15$ represent in context?

50 On the moon, the equation $d = 0.8t^2$ is used to find the distance an object has fallen after t seconds. On Earth, the equation is $d = 4.9t^2$.

- **a** A rock is thrown from a height of 80 metres on Earth. Sketch the graph of the equation on a graphing calculator or other technology to find the time taken to hit the ground to the nearest second.
- **b** A rock is thrown from a height of 80 metres on the Moon. Sketch the graph of the equation on a graphing calculator or other technology to find the time taken to hit the ground to the nearest second.

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